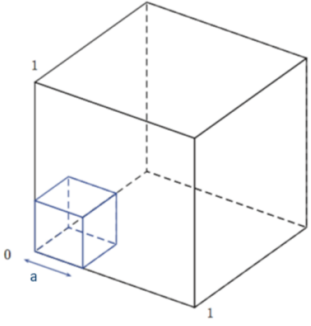
Introduction to Machine Learning

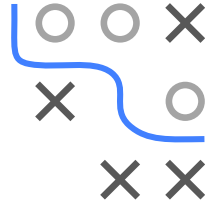
Curse of Dimensionality



Learning goals

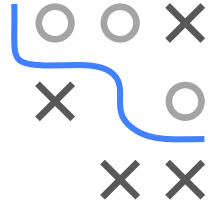
- Understand that our intuition about geometry fails in high-dimensional spaces
- Understand the effects of the curse of dimensionality

CURSE OF DIMENSIONALITY



- The phenomenon of data becoming sparse in high-dimensional spaces is one effect of the **curse of dimensionality**.
- The **curse of dimensionality** refers to various phenomena that arise when analyzing data in high-dimensional spaces that do not occur in low-dimensional spaces.
- Our intuition about the geometry of a space is formed in two and three dimensions.
- We will see: This intuition is often misleading in high-dimensional spaces.

CURSE OF DIMENSIONALITY: EXAMPLE



To illustrate one of the problematic phenomena of data in high dimensional data, we look at an introductory example:

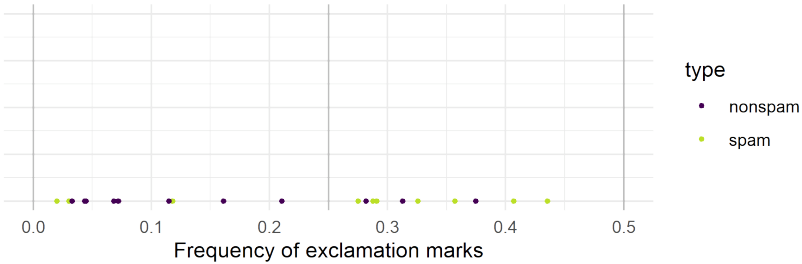
We are given 20 emails, 10 of them are spam and 10 are not. Our goal is to predict if a new incoming mail is spam or not.

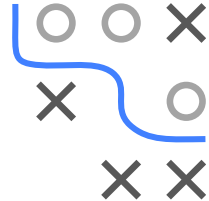
For each email, we extract the following features:

- frequency of exclamation marks (in %)
- the length of the longest sequence of capital letters
- the frequency of certain words, e.g., “free” (in %)
- ...

... and we could extract many more features!

CURSE OF DIMENSIONALITY: EXAMPLE / 2



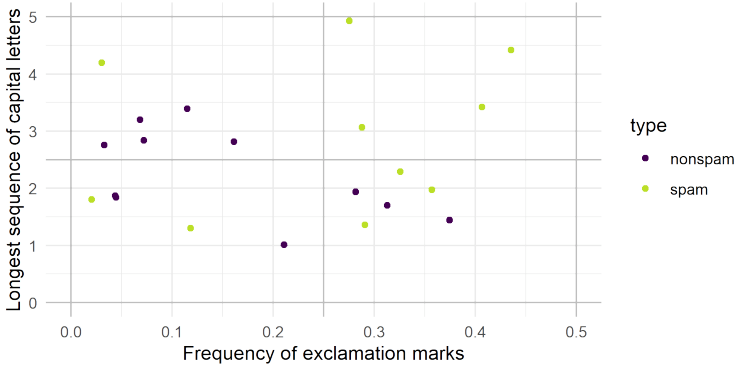


Based on the frequency of exclamation marks, we train a very simple classifier (a decision stump with split point $\mathbf{x} = 0.25$):

- We divide the input space into 2 equally sized regions.
- In the second region $[0.25, 0.5]$, 7 out of 10 are spam.
- Given that at least 0.25% of all letters are exclamation marks, an email is spam with a probability of $\frac{7}{10} = 0.7$.

CURSE OF DIMENSIONALITY: EXAMPLE / 3

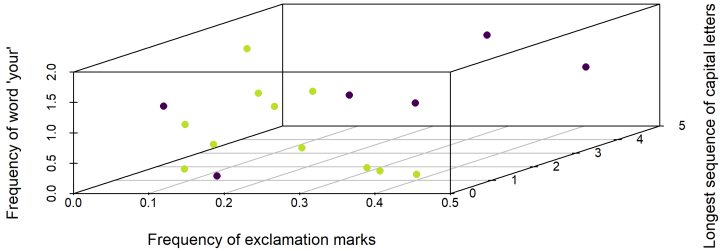
Let us feed more information into our classifier. We include a feature that contains the length of the longest sequence of capital letters.



- In the 1D case we had 20 observations across 2 regions.
- The same number is now spread across 4 regions.

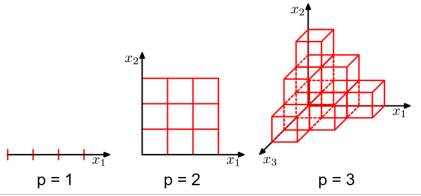
CURSE OF DIMENSIONALITY: EXAMPLE / 4

Let us further increase the dimensionality to 3 by using the frequency of the word “your” in an email.



CURSE OF DIMENSIONALITY: EXAMPLE / 5

- When adding a third dimension, the same number of observations is spread across 8 regions.
- In 4 dimensions the data points are spread across 16 cells, in 5 dimensions across 32 cells and so on ...
- As dimensionality increases, the data become **sparse**; some of the cells become empty.
- There might be too few data in each of the blocks to understand the distribution of the data and to model it.

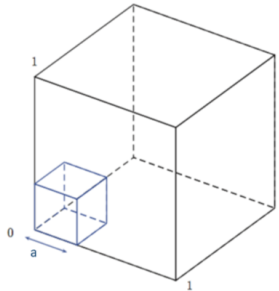


Bishop, Pattern Recognition and Machine Learning, 2006

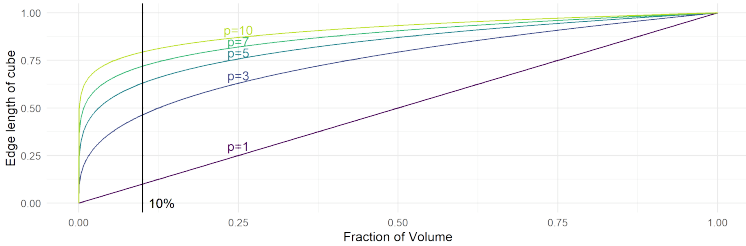
Geometry of High-Dimensional Spaces

THE HIGH-DIMENSIONAL CUBE

- We embed a small cube with edge length *a* inside a unit cube.
- How long does the edge length *a* of this small hypercube have to be so that the hypercube covers $10\%, 20\%, \ldots$ of the volume of the unit cube (volume 1)?



THE HIGH-DIMENSIONAL CUBE / 2



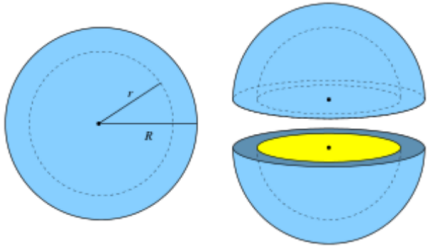
$$a^p = \frac{1}{10} \Leftrightarrow a = \frac{1}{\sqrt[p]{10}}$$

- So: covering 10% of total volume in a cell requires cells with almost 50% of the entire range in 3 dimensions, 80% in 10 dimensions.

THE HIGH-DIMENSIONAL SPHERE

Another manifestation of the **curse of dimensionality** is that the majority of data points are close to the outer edges of the sample. Consider a hypersphere of radius 1. The fraction of volume that lies in the ϵ-"edge", $\epsilon := R - r$, of this hypersphere can be calculated by the formula

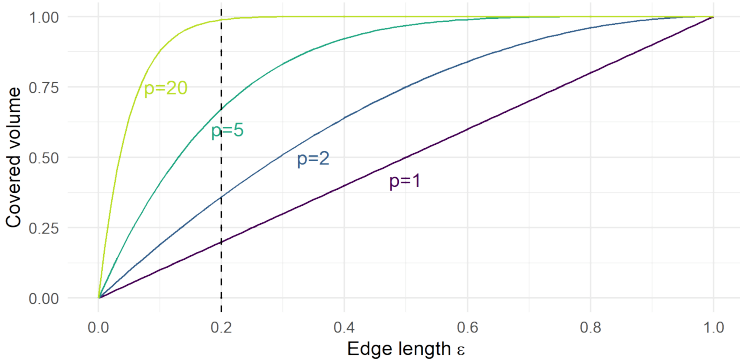
$$1 - \left(1 - \frac{\epsilon}{R}\right)^p .$$



If we peel a high-dimensional orange, there is almost nothing left.

THE HIGH-DIMENSIONAL SPHERE / 2

Consider a 20-dimensional sphere. Nearly all of the volume lies in its outer shell of thickness 0.2:

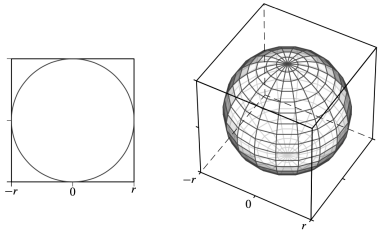


HYPHERSPHERE WITHIN HYPERCUBE

Consider a p-dimensional hypersphere of radius r and volume $S_p(r)$ inscribed in a p-dimensional hypercube with sides of length $2r$ and volume $C_p(r)$. Then it holds that

$$\lim_{p\to\infty} \frac{S_p(r)}{C_p(r)} = \lim_{p\to\infty} \frac{\left(\frac{\pi^{\frac{p}{2}}}{\Gamma(\frac{p}{2}+1)}\right) r^p}{(2r)^p} = \lim_{p\to\infty} \frac{\pi^{\frac{p}{2}}}{2^p\Gamma(\frac{p}{2}+1)} = 0,$$

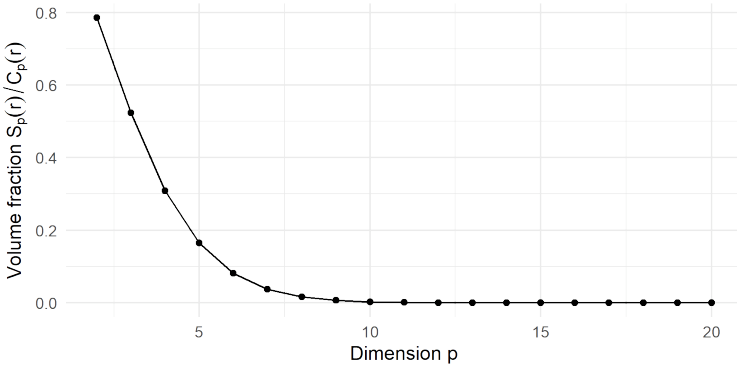
i.e., as the dimensionality increases, most of the volume of the hypercube can be found in its corners.



Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Analysis: Fundamental Concepts and Algorithms, 2014

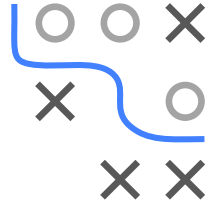
HYPHERSPHERE WITHIN HYPERCUBE / 2

Consider a 10-dimensional sphere inscribed in a 10-dimensional cube. Nearly all of the volume lies in the corners of the cube:



Note: For $r > 0$, the volume fraction $\frac{S_p(r)}{C_p(r)}$ is independent of r.

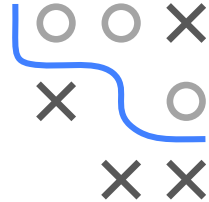
UNIFORMLY DISTRIBUTED DATA



The consequences of the previous results for uniformly distributed data in the high-dimensional hypercube are:

- Most of the data points will lie on the boundary of the space.
- The points will be mainly scattered on the large number of corners of the hypercube, which themselves will become very long spikes.
- Hence the higher the dimensionality, the more similar the minimum and maximum distances between points will become.
- This degrades the effectiveness of most distance functions.
- Neighborhoods of points will not be local anymore.

GAUSSIANS IN HIGH DIMENSIONS



A further manifestation of the **curse of dimensionality** appears if we consider a standard Gaussian $N_p(\mathbf{0}, \boldsymbol{I}_p)$ in p dimensions.

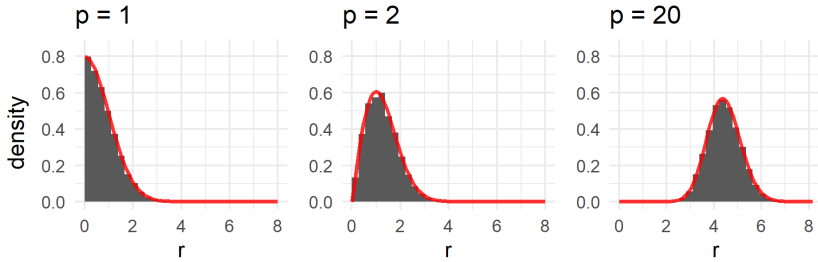
- After transforming from Cartesian to polar coordinates and integrating out the directional variables, we obtain an expression for the density $p(r)$ as a function of the radius r (i.e., the point's distance from the origin), s.t.

$$p(r) = \frac{S_p r^{p-1}}{(2\pi\sigma^2)^{p/2}} \exp\left(-\frac{r^2}{2\sigma^2}\right),$$

  where S_p is the surface area of the p-dimensional unit hypersphere.
- Thus $p(r)\delta r$ is the approximate probability mass inside a thin shell of thickness δr located at radius r.

GAUSSIANS IN HIGH DIMENSIONS / 2

- To verify this functional relationship empirically, we draw 10^4 points from the p-dimensional standard normal distribution and plot $p(r)$ over the histogram of the points' distances to the origin:



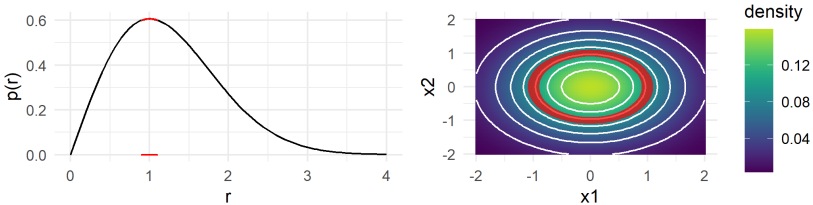
- We can see that for large p the probability mass of the Gaussian is concentrated in a fairly thin "shell" rather far away from the origin. This may seem counterintuitive, but:

GAUSSIANS IN HIGH DIMENSIONS / 3

- For the probability mass of a hyperspherical shell it follows that

$$\int_{r-\frac{\delta r}{2}}^{r+\frac{\delta r}{2}} p(\tilde{r})d\tilde{r} = \int_{r-\frac{\delta r}{2} \leq ||\mathbf{x}||_2 \leq r+\frac{\delta r}{2}} f_p(\tilde{\mathbf{x}})d\tilde{\mathbf{x}},$$

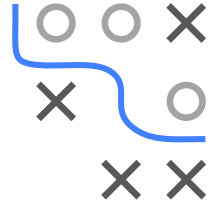
where $f_p(\mathbf{x})$ is the density of the p-dimensional standard normal distribution and $p(r)$ the associated radial density.



Example: 2D normal distribution

- While f_p becomes smaller with increasing r, the region of the integral -the hyperspherical shell- becomes bigger.

INTERMEDIATE REMARKS



However, we can find effective techniques applicable to high-dimensional spaces if we exploit these properties of real data:

- Often the data is restricted to a manifold of a lower dimension. (Or at least the directions in the feature space over which significant changes in the target variables occur may be confined.)
- At least locally small changes in the input variables usually will result in small changes in the target variables.